PACE UM PU COLLEGE

I PUC 26-27 JEE WT- 31-10-2026

31 Oct 2026

Student ID :
Total Marks : 300
Duration : 180

Maths

1) Let α, β be the roots of $x^2+x+1=0$. The equation whose roots are α^{25} *and* β^{22}

(1) $x^2+x-1=0$ (2) $x^2-x+1=0$

(3) $x^2+x+1=0$ (4) None of these

2) If 12 and 9 are respectively A.M and G.M of two numbers. Then the numbers are the roots of the equation

(1) $x^2-24x+81=0$ (2) $x^2+24x-81=0$

(3) $x^2-81x+24=0$ (4) $x^2+24x+81=0$

3) If α,β are the roots of $2x^2-x+7=0$ then the equation whose roots are $2-3\alpha$ and $2-3\beta$ is

(1) $2x^2-5x+65=0$ (2) $2x^2+5x+65=0$

(3) $2x^2-5x-65=0$ (4) $2x^2+5x-65=0$

4) If the roots of $\left(a^2+b^2\right)x^2+2\left(bc+ad\right)x+\left(c^2+d^2\right)=0$ are equal then

(1) $\frac{a}{b}=\frac{d}{c}$ (2) $\frac{a}{c}+\frac{b}{d}=0$

(3) $\frac{a}{c}=\frac{b}{d}$ (4) a+b=c+d

5) If 2 lies between the roots of $x^2+\left(a+1\right)x-a-2=0$ then $a\in$

(1) $(4,\infty)$ (2) $(-4,\infty)$

(3) $(-4,4)$ (4) $(-\infty,-4)$

6) If one root of $\left(a^2-5a+3\right)x^2+\left(3a-1\right)x+2=0$ is twice the other, then a=

(1) $\dfrac{2}{3}$ (2) $-\dfrac{2}{3}$ (3) $\dfrac{1}{3}$ (4) $-\dfrac{4}{3}$

7) Let f(x)=ax^2+bx+c ,a,b $\in R$,a$\neq 0$ satisfying f(1)+f(2)=0. Then the equation f(x) =0 has

(1) no real root (2) 1 and 2 are real roots

(3) two distinct roots (4) two equal roots

8) The number of values of a such that $x^2-11x+a=0$ and $x^2-14x+2a=0$ have a common root

(1) 0 (2) 2 (3) 3 (4) 4

9) If a $\in$ Z and the equation (x-3)(x-a)+1=0 has integral roots, then the value of 'a' are

(1) 2 or 5 (2) 1 or 5 (3) 1 or 4 (4) 2 or 4

10) If the quadratic equation $ax^2-2cx+b=0$ and $ax^2-2bx+c=0$(b $\neq$ c) have a common root, then a+4b+4c is equal to

(1) -2 (2) -1 (3) 0 (4) 1

11) If α,β are the roots of $ax^2+bx+c=0$ and $A_n=\alpha^n+\beta^n$, then ,$aA_{n+2}+bA_{n+1}+cA_n=$

(1) 0 (2) 1 (3) a+b+c (4) abc

12) Real roots of the equation $x^2+6\left|x\right|+5=0$ are

(1) -1,-5 (2) 1,5 (3) 1,-1 (4) no real roots

13) The sum of the roots of the equation $\left|x^2+4x+3\right|+2x+5=0$ is

(1) $5+\sqrt{3}$ (2) $5-\sqrt{3}$

(3) $-5+\sqrt{3}$ (4) $-5-\sqrt{3}$

14) The sum of the real roots of the equation $\left|x-2\right|^2+\left|x-2\right|-2=0$ is

(1) 0 (2) 8 (3) -2 (4) 4

15) If $x\in R$ the expression $\frac{x^2+2x-11}{x-3}$does not take value in

(1) $(-\infty,4)$ (2) $(12,\infty)$

(3) R-(4,12) (4) (4,12)

16) The sum of distinct values of x such that $\left|x^2-x-6\right|=x+2$ is

(1) 0 (2) 2 (3) -2 (4) 4

17) If the roots of the equation $x^2-2ax+a^2+a-3=0$ are real and less than 3, then

(1) a<2 (2) $2\le a\le 3$ (3) $3<a\le 4$ (4) a>4

18) If and $\alpha\neq\beta$ and $\alpha^2=5\alpha-3,\beta^2=5\beta-3$ then the equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

(1) $3x^2+19x+3=0$ (2) $3x^2-19x+3=0$

(3) $3x^2-19x-3=0$ (4) $3x^2-16x+1=0$

19) If 2a+3b+6c=0,a,b,c$\in R$ Then the equation $ax^2+bx+c=0$ has a root in

(1) (0,1) (2) (2,3) (3) (4,5) (4) None of these

20) If x is real, the maximum value of $\frac{3x^2+9x+17}{3x^2+9x+7}$is

(1) $\frac{1}{4}$ (2) 41 (3) 1 (4) $\frac{17}{7}$

21) Let 'n' be the number of quadratic equations of the form $x^2+bx+c=0$, where b, c can take values from {0,1,2.....,9} such that it has no roots which are equal and opposite sign. Then the value of 'n' is

22) Let $x^2-3x+1=0$, then $\frac{x^{14}+1}{x^7}$=

23) If α,β are the roots of $4x^2-16x+c=0$, c>0 such that ,1<α <2<β<3, then the number of integral values of 'c' is

24) If α,β be the roots of $x^2+x+2=0$ and γ,δ be the roots of $x^2+3x+4=0$ then $\left(\alpha+\gamma\right)\left(\alpha+\delta\right)\left(\beta+\gamma\right)\left(\beta+\delta\right)=$

25) If one root of the equation $5x^2+13x+k=0$ is reciprocal of other , then the value of k is

Physics

26) Two identical particles, each of mass m are placed at two points on the x-axis. If both the particles are moving parallel to x-axis towards +x direction with velocity 2 m/s, the velocity of CM of the system is

(1) 2 m/s (2) 0 m/s (3) -2 m/s (4) 4 m/s

27) Two particles of mass 1 kg and 0.5 kg are moving in the same direction with speed of 2m/s and 6 m/s respectively on a smooth horizontal surface. The speed of centre of mass of the system is

(1) $\frac{10}{3}$ m/s (2) $\frac{10}{7}$ m/s

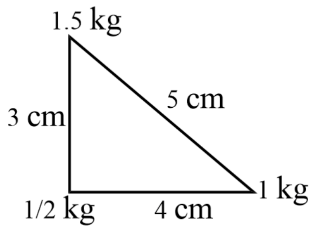
(3) $\frac{11}{2}$ m/s (4) $\frac{12}{3}$ m/s

28) The distance between the centres of carbon and oxygen atoms in the carbon monoxide gas molecule is $1.13 \times 10^{-10} m$. The distance of centre of mass of the molecule relative to carbon atom is

(1) $0.48 \times 10^{-10} m$ (2) $0.64 \times 10^{-10} m$

(3) $0.56 \times 10^{-10} m$ (4) $0.36 \times 10^{-10} m$

29) The coordinates of the centre of the mass of the system of the given figure



(1) (1.3, 1.5) (2) (2.5, 3.2)

(3) (0.8, 1.4) (4) (0, 0)

30) A ladder rests against a frictionless vertical wall with its upper end 6 m above the ground and the lower end 4 m away from the wall. The weight of the ladder is 500 N and its C.G. is at 1/3rd distance from the lower end. The wall's reaction will be: (in Newton)

(1) 111 (2) 333 (3) 222 (4) 129

31) Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2, respectively, about their axes is $I_1 : I_2 = 1 : 16$ then the value of α is

(1) $\sqrt{2}$ (2) 2 (3) 4 (4) $2\sqrt{2}$

32) The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at (2, 0, - 3), about the point (2, - 2, - 2), is given by:

(1) $-7\hat{i} - 8\hat{j} - 4\hat{k}$ (2) $-4\hat{i} - \hat{j} - 8\hat{k}$

(3) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ (4) $-7\hat{i} - 4\hat{j} - 8\hat{k}$

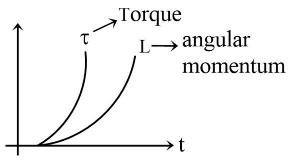
33) Six identical particles each of mass m are arranged at the corners of a regular hexagon of side length a. If the mass of one of the particle is doubled, the shift in the centre of mass is

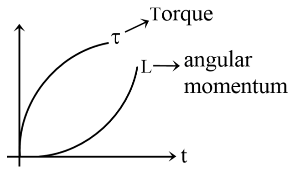
(1) a (2) $\frac{6a}{7}$ (3) $\frac{a}{7}$ (4) $\frac{a}{\sqrt{3}}$

34) Two spherical bodies of mass M and 5M and radii R and 2R respectively are released in free space with initial separation between their centres equal to 12 R. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before the collision is

(1) 2.5 R (2) 4.5 R (3) 7.5 R (4) 1.5 R

35) A particle is projected with initial velocity u at an angle with the horizontal. The variation of torque and angular momentum of particle about the point of projection with time, will be:

(1)


(2)


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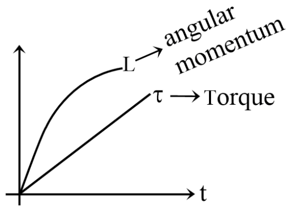

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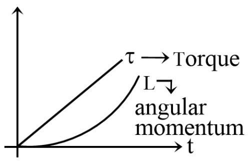
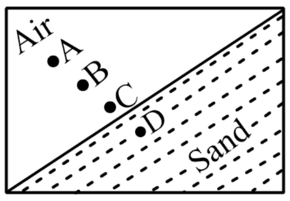


36) Which of the points is the likely position of the centre of mass of the system shown in the figure?



(1) A (2) D (3) B (4) C

37) A body of mass $m_1 = 4$ kg moves at 5i m/s and another body of mass $m_2 = 2$ kg moves at 10i m/s. The kinetic energy of centre of mass is:

(1) $\frac{200}{3}J$ (2) $\frac{500}{3}J$ (3) $\frac{400}{3}J$ (4) $\frac{800}{3}J$

38) Two particles of masses M and 4M are released from a distance 10 metres from each other. Find the point of collision from smaller particle.

(1) 8 metres (2) 2 metres

(3) 5 metres (4) 6 metres

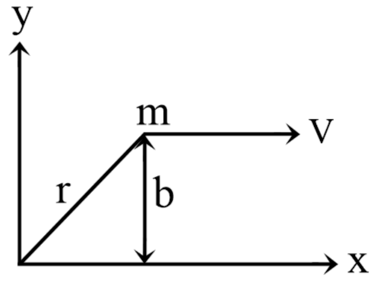
39) A body weighs 8 g when placed in one pan and 18 g when placed on the other pan of a false balance. If the beam is horizontal when both the pans are empty, the true weight of the body is:

(1) 13 g (2) 9 g (3) 22 g (4) 12 g

40) What is the distance of the centre of mass of a half-ring from centre if the ring has a radius of 0.5 m?

(1) $\frac{1}{\pi}$ (2) $\frac{1}{3\pi}$ (3) $\frac{2}{3\pi}$ (4) $\frac{1}{2\pi}$

41) If a particle of mass 'm' is moving with constant velocity 'v' parallel to x-axis in the x-y plane as shown in the figure. Its angular momentum with respect to origin at any time 't' will be



(1) $mvb\hat{k}$ (2) $-mvb\hat{k}$ (3) $mvb\hat{i}$ (4) $mv\hat{i}$

42) A uniform thin bar of mass 6 m and length '12 L' is bent to make a regular hexagon. Its moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of the hexagon is

(1) $20\,mL^2$ (2) $6\,mL^2$

(3) $\frac{20}{3}mL^2$ (4) $\frac{10}{3}mL^2$

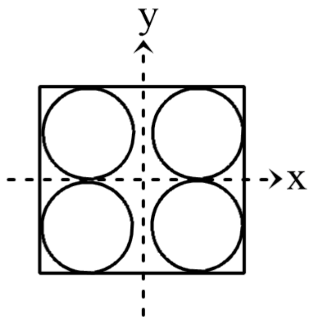
43) The moment of inertia of a uniform circular discabout its central axis perpendicular to the plane is 'I'. Its M.I about a tangent in its plane is equal to

(1) 2I (2) 2.5 I (3) 1.5 I (4) $\frac{I}{2}$

44) Consider a uniform square plate of side 'a' and mass 'm'. The moment of inertia of this plate about an axis perpendicular to its plane and passing through one of its corners is

(1) $\frac{5}{6}ma^2$ (2) $\frac{1}{12}ma^2$ (3) $\frac{7}{12}ma^2$ (4) $\frac{2}{3}ma^2$

45) Four holes of radius R are cut from a thin square plate of side 4R and mass M. The moment of inertia of the remaining portion about z-axis is



(1) $\frac{\pi}{2}MR^2$ (2) $\left(\frac{4}{3} - \frac{\pi}{4}\right)MR^2$

(3) $\left(\frac{8}{3} - \frac{10\pi}{16}\right)MR^2$ (4) $\left(\frac{4}{3} - \frac{\pi}{16}\right)MR^2$

46) A wheel starts rotating from rest and attains an angular velocity of 60 rad/sec in 5 seconds. The total angular displacement in radians will be-

47) The radius of gyration of a body about an axis at a distance of 4 cm from its centre of mass is 5 cm. The radius of gyration about a parallel axis through the centre of mass is

(Note : The answer should be cm)

48) Four particles each of mass 1 kg are at four corners of a square of side 1 m. The M.I of the system about a normal axis through centre of square is

(Note : The answer should be kgm^2)

49) A solid cylinder of mass 20 kg and radius 20 cm rotates about its axis with an angular speed 100 rad s^{-1}. The angular momentum of the cylinder about its axis is

(Note : The answer should be Js)

50) A particle of mass 2 kg moves in space such that its position vector varies as $\vec{r} = 2t\hat{i} + 3t^2\hat{j}$. The angular momentum of the particle about the origin at t = 2 s is:

(Note : The answer should be $\hat{k}$)

Chemistry

51) resonance effect involves:

(1) Delocalization of lone pair along a conjugated system.

(2) Delocalization of pi-electrons along a conjugated system.

(3) Delocalization of negative charge along a conjugated system.

(4) All are correct.

52) In which of the following first resonating structure is more stable than the second?

(1) $C\underset{2}{\overset{\ominus}{H}} - CH = O \leftrightarrow CH_2 = CH - O^{\ominus}$

(2) $\overset{\oplus}{C}H_2 - O - CH_3 \leftrightarrow CH_2 = \overset{\oplus}{O} - CH_3$

(3) $CH_2 = CH - NH_2 \leftrightarrow C\overset{\ominus}{H_2} - CH = \overset{\oplus}{N}H_2$

(4) $\overset{\oplus}{O} - CH = CH_2 \leftrightarrow O = CH - \overset{\oplus}{C}H_2$

53) The hyper conjugative stabilities of tertiary – butyl carbocation and but -2-ene, respectively, are due to

(1) $\sigma \to p$ (empty) and $\sigma \to \pi^*$ electron delocalisations

(2) $\sigma \to \sigma^*$ and $\sigma \to \pi$ electron delocalisations

(3) $\sigma \to p$ (filled) and $\sigma \to \pi$ electron delocalisations

(4) p (filled) $\to \sigma^*$ and $\sigma \to \pi^*$ electron delocalisations

54) Which of the following has the shortest C - Cl bond?

(1) $Cl - CH = CH - OCH_3$

(2) $Cl - CH = CH - CH_3$

(3) $Cl - CH = CH_2$

(4) $Cl - CH = CH - NO_2$

55) The correct stability order of alkenes on the basis of hyper conjugation is

(1) Ethylene > Propene > 2 – butane > 2, 3 – dimethyl -2 - butene

(2) 2- butene > 2,3 – dimethyl -2 – butene > Ethylene > propene

(3) 2,3 – dimethyl -2- butene > 2 – butene > propene > ethylene

(4) 2,3 – dimethyl -2- butene > 2 – butene > ethylene > propene

56) Which of the following order is correct regarding the -I effect of the substituents?

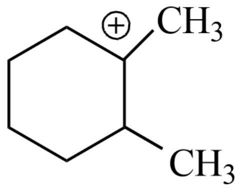
(1) $-NR_2 > -OR > -F$

(2) $-NR_2 < -OR > -F$

(3) $-NR_2 > -OR < -F$

(4) $-NR_2 < -OR < -F$

57) How many hyper conjugable H-atoms does this cation has?



(1) 0 (2) 6 (3) 12 (4) 15

58) The stability of $Me_2C = CH_2$ is more than that of $MeCH_2CH = CH_2$ due to

(1) inductive effect of the Me group

(2) resonance effect of the Me group

(3) hyperconjugative effect of the Me group

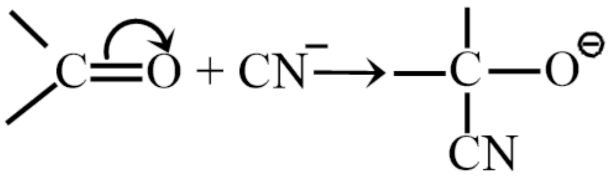
(4) resonance as well as inductive effect of the Me group

59) According to hyper conjugation the most stable alkene is

(1) Ethylene (2) 2,3 – dimethyl – 2 – butene

(3) 2 – methyl – 2 – butene (4) 2 – methyl propene

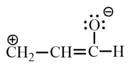
60) The type of electron effect involved in the following reaction

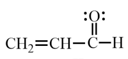


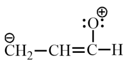
(1) Electromeric effect (2) Inductive effect

(3) Hyper conjugation (4) All the above

61) The correct stability order of above resonating structures is


(I)


(II)


(III)

(1) I > II > III (2) II > I > III

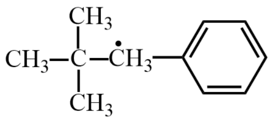
(3) III > II > I (4) II > III > I

62) The most stable carbonium ion is

(1) $(CH_3)_2CH^{\oplus}$ (2) $CH_3CH_2^{\oplus}$

(3) $(CH_3)_3C\text{-}\overset{\oplus}{C}H_2$ (4) $(CH_3)_3\overset{\oplus}{C}$

63) Hyperconjugation occurs in: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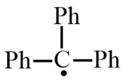
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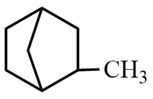
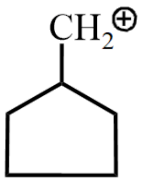
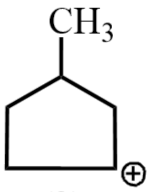

(III)

(1) I only (2) II only (3) III only (4) I and III

64) Decreasing order of stability of following carbocations is


(1)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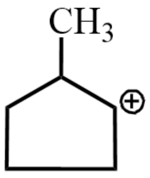
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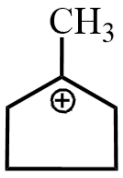

(4)

(1) 4 > 2 > 3 > 1 (2) 2 > 4 > 3 > 1

(3) 3 > 1 > 4 > 2 (4) 4 > 2 > 1 > 3

65) Which one of the following resonating structures of 1-methoxy -1, 3-butadiene is least stable?

(1) $\overset{\ominus}{C}H_2 - CH = CH - CH = \overset{\oplus}{O} - CH_3$

(2) $CH_2 = CH - \overset{\ominus}{C}H - CH = \overset{\oplus}{O} - CH_3$

(3) $\overset{\ominus}{C}H_2 - \overset{\oplus}{C}H - CH = CH - O - CH_3$

(4) $CH_2 = CH - \overset{\ominus}{C}H - \overset{\oplus}{C}H - O - CH_3$

66) Which one of the following series contain electrophiles only?

(1) $AlCl_3$, SO_3, $\overset{\oplus}{N}O_2$ (2) H_2O, SO_3, H_3O^+

(3) NH_3, H_2O, $AlCl_3$ (4) H_2O, $\overset{\oplus}{C}l$, NH_3

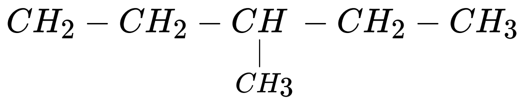
67) Which of the following behave like both electrophile as well as nucleophile

A) CH_3CN B) CH_4

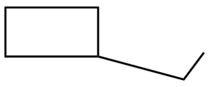
C) N_2 D) $H_2C = O$

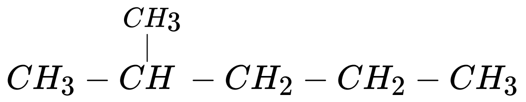
(1) A,B,C (2) A,D (3) B,C (4) A,B,C,D

68) Among the following, identify the compound that is not an isomer of hexane

(1)

$CH_2 - CH_2 - \underset{CH_3}{\underset{|}{CH}} - CH_2 - CH_3$

(2) $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CH_3$

(3)


(4)

$CH_3 - \overset{CH_3}{\overset{|}{CH}} - CH_2 - CH_2 - CH_3$

69) Match LIST-I with LIST-II.

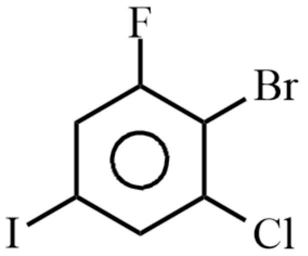
	List I Isomeric pairs		List II Type of isomers
A	Propanamine and N-Methylethanamine	I.	Metamers
B	Hexan-2-one and Hexan-3-one	II.	Positional isomers
C	Ethanamide and Hydroxyethanimine	III.	Functional isomers
D	o-ntrophenol and p-nitrophenol	IV.	Tautomers

Choose the correct answer from the options given below.

(1) A-III, B-I, C-IV, D-II (2) A-IV, B-III, C-I, D-II

(3) A-II, B-III, C-I, D-IV (4) A-III, B-IV, C-I, D-II

70) The correct name of



is

(1) 1-Bromo-6-chloro-2-fluoro-4-iodobenzene

(2) 1-Bromo-2-chloro-6-fluoro-4-iodobenzene

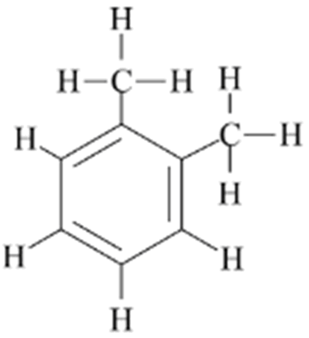
(3) 2-Bromo-1-chloro-3-fluoro-5-iodobenzene

(4) 2-Bromo-3-chloro-5-ido - 1- fluoro benzene

71) Number of metameric 2^0 amines of molecular formula $C_4H_{11}N$ is:

72) The number of structural isomeric compounds possible with molecular formula $C_3H_6Cl_2$

73) The number of σ bonds in *o*-xylene is



74) Number of structural isomers for C_5H_{12} is

75) If the energies of Two resonating structures of a molecule are 71 kJ mol^{-1} and 62 kJ mol^{-1} and the actual energy of the substance is 38 kJ mol^{-1}, the resonance energy of the substance would be (Note : The answer should be $kJ\,mol^{-1}$)